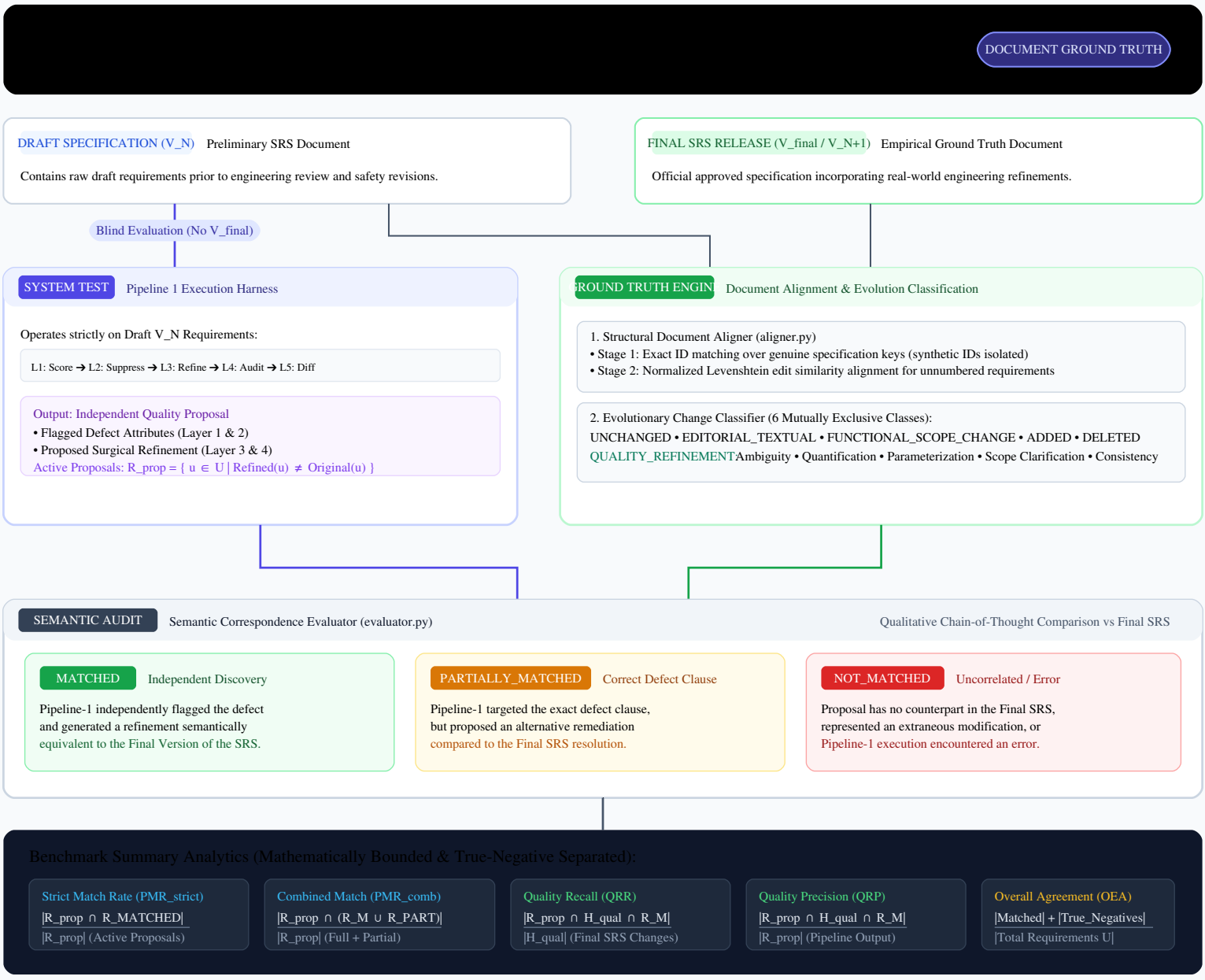

DOCUMENT GROUND TRUTH
DRAFT SPECIFICATION (V_N)
Preliminary SRS Document
Contains raw draft requirements prior to engineering review and safety revisions.
FINAL SRS RELEASE (V_final / V_N+1)
Empirical Ground Truth Document
Official approved specification incorporating real-world engineering refinements.
Blind Evaluation (No V_final)
SYSTEM TEST
Pipeline 1 Execution Harness
Operates strictly on Draft V_N Requirements:
L1: Score ➔ L2: Suppress ➔ L3: Refine ➔ L4: Audit ➔ L5: Diff
Output: Independent Quality Proposal
• Flagged Defect Attributes (Layer 1 & 2)
• Proposed Surgical Refinement (Layer 3 & 4)
Active Proposals: R_prop = { u ∈ U | Refined(u) ≠ Original(u) }
ROUND TRUTH ENGIN
Document Alignment & Evolution Classification
1. Structural Document Aligner (aligner.py)
• Stage 1: Exact ID matching over genuine specification keys (synthetic IDs isolated)
• Stage 2: Normalized Levenshtein edit similarity alignment for unnumbered requirements
2. Evolutionary Change Classifier (6 Mutually Exclusive Classes):
UNCHANGED • EDITORIAL_TEXTUAL • FUNCTIONAL_SCOPE_CHANGE • ADDED • DELETED
QUALITY_REFINEMENTAmbiguity • Quantification • Parameterization • Scope Clarification • Consistency
SEMANTIC AUDIT
Semantic Correspondence Evaluator (evaluator.py)
Qualitative Chain-of-Thought Comparison vs Final SRS
MATCHED
Independent Discovery
Pipeline-1 independently flagged the defect
and generated a refinement semantically
equivalent to the Final Version of the SRS.
PARTIALLY_MATCHED
Correct Defect Clause
Pipeline-1 targeted the exact defect clause,
but proposed an alternative remediation
compared to the Final SRS resolution.
NOT_MATCHED
Uncorrelated / Error
Proposal has no counterpart in the Final SRS,
represented an extraneous modification, or
Pipeline-1 execution encountered an error.
Benchmark Summary Analytics (Mathematically Bounded & True-Negative Separated):
Strict Match Rate (PMR_strict)
|R_prop ∩ R_MATCHED|
|R_prop| (Active Proposals)
Combined Match (PMR_comb)
|R_prop ∩ (R_M ∪ R_PART)|
|R_prop| (Full + Partial)
Quality Recall (QRR)
|R_prop ∩ H_qual ∩ R_M|
|H_qual| (Final SRS Changes)
Quality Precision (QRP)
|R_prop ∩ H_qual ∩ R_M|
|R_prop| (Pipeline Output)
Overall Agreement (OEA)
|Matched| + |True_Negatives|
|Total Requirements U|